

1 Midterm

Exercise 1.1. Let A be a simple associative algebra. Show that A^+ is a simple Jordan algebra.

Proof. Let I be an ideal of $A^{(+)}$. $\forall x, y \in I$ ($x = y$ is allowed so we can always pick elements in this way). $\forall a \in A$.

$$[a, x \circ y] = x \circ [a, y] - [x, a] \circ y \in I$$

Since $x \circ y \in I$, $a \circ (x \circ y) \in I$. Hence

$$2a(x \circ y) = [a, x \circ y] + a \circ (x \circ y) \in I$$

Now $\forall b \in A$, $(2a(x \circ y)) \circ b \in I$. Since $b((2a(x \circ y))) = 2ba(x \circ y)$. Recall in the last step, a is arbitrary in A , we have $2ba(x \circ y) \in I$. Hence

$$2a(x \circ y)b = (2a(x \circ y)) \circ b - b((2a(x \circ y))) \in I$$

Since a, b are arbitrary in A . Denote $(x \circ y) = z$. Rewrite it into $2AzA \subseteq I$.

Now $2AzA$ is an ideal of A . If $2AzA = A$, then $A = 2AzA \subseteq I \subseteq A^{(+)} \subseteq A$. Hence $I = A^{(+)}$.

If $2AzA = 0$, since $\text{char} \neq 2$, $AzA = 0$. Then Az and zA are two double sided ideal of A . But z can not be 1. Since If $z = 1$ then $AzA = A^2 = 0$ contradict to A is simple. (If $A^2 = 0$ then $\forall a \in A$, $\mathbb{Z}a$ is a double sided ideal of A).

Then it forces $z = 0$. If not, $\mathbb{Z}z$ is a double sided ideal of A .

Conclusion: we now show that if $\exists x, y \in I$ such that $x \circ y \neq 0$ then AzA can not be 0 and hence $AzA = I = A^{(+)}$. We want to show in the second case $\forall x, y \in I$, $x \circ y = 0$ implies $I = 0$.

Suppose $\exists 0 \neq x \in I$. $\forall a \in A$, $x \circ (x \circ a) = 0$. i.e. $x^2a + ax^2 + 2xax = 0$. since $x \circ x = 2x^2 = 0$ and $\text{char} \neq 2$ forces $x^2 = 0$. Hence $2xax = 0$, i.e. $xax = 0$, in other words $xAx = 0$. Then $(AxA)^2 \subseteq AxAxA = 0$. Since AxA is an ideal, $AxA = A$ or $AxA = 0$, but $A^2 \neq 0$ as we discussed before. Hence $AxA = 0, \forall x \in I$. And as we talked before it forces $x = 0$. Thus, $I = 0$.

Therefore, $I = A^{(+)}$ or $I = 0$, i.e. $A^{(+)}$ is a simple Jordan algebra. $\square$

Exercise 1.2. Let A be a commutative associative algebra with a derivation D satisfying $D^3 = 0$. Define a new multiplication in A

$$a \star b = D(a)D(b).$$

Show that $(A, \star)$ is a Jordan algebra.

Proof. Since A is commutative and D is linear, we have

$$a \star b = D(a)D(b) = D(b)D(a) = b \star a,$$

and $\star$ is bilinear.

Let $x, y \in A$. Write products in A multiplicatively and use that D is a derivation:

$$D(uv) = D(u)v + uD(v).$$

Compute the right-hand side:

$$\begin{aligned} (x \star x) \star (y \star x) &= D(x \star x) \cdot D(y \star x) \\ &= D(D(x)^2) \cdot D(D(y)D(x)) \\ &= [2D(x)D^2(x)] \cdot [D^2(y)D(x) + D(y)D^2(x)] \\ &= 2D(x)^2D^2(x)D^2(y) + 2D(x)[D^2(x)]^2D(y). \end{aligned}$$

Compute the left-hand side:

$$\begin{aligned}
((x \star x) \star y) \star x &= D(((x \star x) \star y)) \cdot D(x) \\
&= D(D(x \star x) \cdot D(y)) \cdot D(x) \\
&= D(2D(x)D^2(x)D(y)) \cdot D(x) \\
&= 2D(D(x)D^2(x)D(y)) \cdot D(x).
\end{aligned}$$

Using $D(uvw) = D(u)vw + uD(v)w + uvD(w)$ with $u = D(x)$, $v = D^2(x)$, $w = D(y)$, and $D^3(x) = 0$, we get

$$\begin{aligned}
D(uvw) &= D^2(x) \cdot D^2(x) \cdot D(y) + D(x) \cdot 0 \cdot D(y) + D(x) \cdot D^2(x) \cdot D^2(y) \\
&= [D^2(x)]^2 D(y) + D(x)D^2(x)D^2(y).
\end{aligned}$$

Hence

$$\begin{aligned}
((x \star x) \star y) \star x &= 2([D^2(x)]^2 D(y) + D(x)D^2(x)D^2(y)) \cdot D(x) \\
&= 2D(x)[D^2(x)]^2 D(y) + 2D(x)^2 D^2(x)D^2(y).
\end{aligned}$$

This matches the right-hand side exactly, so the Jordan identity holds.

Therefore $(A, \star)$ is a commutative algebra satisfying the Jordan identity, hence a Jordan algebra. $\square$

Exercise 1.3. Let J be a Jordan algebra. Show that for any $a, b \in J$ the operator $[R_a, R_b]$ is a derivation of J .

Proof. For any algebra A , Let D be a linear operator $D : A \rightarrow A$. D is a derivation if and only if

$$D(ab) = D(a)b + aD(b), \forall a, b \in A$$

Write it into operators on b we have $[R_a, D] = R_a D, \forall a \in A$.

Recall linearization:

$$R_{(bc)a} - R_{(ac)b} + R_b R_a R_c + R_c R_a R_b - R_a R_b R_c - R_c R_b R_a = 0$$

i.e.

$$R_{(bc)a} - R_{(ac)b} + [R_b, R_a]R_c + R_c[R_a, R_b] = 0$$

i.e.

$$R_{(a,c,b)} + [[R_a, R_b], R_c] = 0$$

Or

$$R_{(a,c,b)} = [R_c, [R_a, R_b]] \tag{IV}$$

We can write $(a, c, b) = (ac)b - a(cb) = cR_a R_b - cR_b R_a = c[R_a, R_b]$. Thus we have $[R_c, [R_a, R_b]] = R_{(a,c,b)} = R_{c[R_a, R_b]}$. Hence $[R_a, R_b]$ is a derivation. $\square$

Exercise 1.4. Let J be a Jordan algebra and let e be an idempotent in J . Show that U_e is a projection on J_1 , where J_1 is the Peirce decomposition component with respect to e .

Proof. First, we show that the image of U_e is J_1 . Let $a \in \text{Im}(U_e)$, so $a = bU_e$ for some $b \in J$. By definition, $bU_e = 2(be)e - e^2b = 2(be)e - be$. We check if $ae = a$:

$$ae = (2(be)e - be)e = 2((be)e)e - (be)e$$

$2((be)e)e = 2(be)(ee) + 2(be, e, e)$ where $(be, e, e) = -(e, e, b) - (be, e, e)$ hence $(be, e, e) = -\frac{1}{2}(e, e, b)$. We have $2(be)(ee) + 2(be, e, e) = 2(be)e - (e, e, b) = 2(be)e - eb + e(eb) = 3(be)e - be$. Thus,

$$\begin{aligned} ae &= (3(be)e - be) - (be)e \\ &= 2(be)e - be \\ &= a \end{aligned}$$

Since $ae = a$, a is in J_1 . Thus, $\text{Im}(U_e) \subseteq J_1$.

Conversely, let $a \in J_1$. By definition, $ae = a$. We apply U_e :

$$aU_e = 2(ea)e - e^2a = 2ae - ea = 2a - a = a.$$

This shows that $a \in \text{Im}(U_e)$. Thus, $J_1 \subseteq \text{Im}(U_e)$. Combining both inclusions, we have $\text{Im}(U_e) = J_1$.

Moreover, for any $a \in J$, let $b = aU_e$. We have shown $b \in J_1$. Applying U_e again, $bU_e = b$. So, $aU_eU_e = aU_e$, which means $U_e^2 = U_e$. Thus, U_e is a projection operator.

Next, we find the kernel of U_e . Let $t \in J_0$. By definition, $te = 0$. Then $tU_e = 2(te)e - e^2t = 0$. So, $J_0 \subseteq \ker(U_e)$.

Let $t \in J_{1/2}$. By definition, $te = \frac{1}{2}t$. Then

$$tU_e = 2(te)e - e^2t = 0.$$

So, $J_{1/2} \subseteq \ker(U_e)$.

Therefore, $J_0 \oplus J_{1/2} \subseteq \ker(U_e)$. Since the image is J_1 and $J = J_1 \oplus J_{1/2} \oplus J_0$, the kernel must be exactly $J_0 \oplus J_{1/2}$. $\square$

Exercise 1.5. Let J be a Jordan algebra with a unit element. Define an invertible element in J . Prove that a is invertible in J if and only if U_a is invertible in $\text{Mult}J$.

Proof. Let a be invertible with inverse b . Then

$$\begin{aligned} bU_a &= 2(ba)a - ba^2 = 2a - a = a \\ b^2U_a &= 2(b^2a)a - b^2a^2 = 2ba - 1 = 1 \end{aligned}$$

Observe that $U_1 = \text{Id}_J$, the identity operator in J . Then

$$\text{Id}_J = U_1 = U_{U_a(b^2)} = U_aU_{b^2}U_a$$

this means that U_a is invertible.

Conversely, suppose that U_a is invertible and let $b = aU_a^{-1}$. Then $bU_a = a$. Since $[U_a, R_a] = 0$ we also have $[U_a^{-1}, R_a] = 0$. ($[U_a^{-1}, R_a] = U_a^{-1}[R_a, U_a]U_a^{-1} = 0$). Then

$$\begin{aligned} ba &= aU_a^{-1}R_a = aR_aU_a^{-1} = a^2U_a^{-1} \\ &= 1U_aU_a^{-1} = 1 \end{aligned}$$

and

$$a = bU_a = 2(ba)a - ba^2 = 2a - ba^2.$$

Thus $ba^2 = a \implies a$ is invertible. $\square$

Exercise 1.6. Let J be a Jordan algebra. Show that if $(\text{Mult}(J))^n = 0$ then $J^{2n-1} = 0$ ($n \geq 1$).

Proof. Consider a word $x_1, \dots, x_{2n-1} \in J^{2n-1}$. It may have arbitrary order of operations. For example, when $n = 2$, $abcd$ or $ab(cd)$ or $a((bc)d)$ and so on. All we know is the “length” of this word is $2n - 1$. We rewrite it into:

$$x_i R_{x_1, \dots, \hat{x}_i, \dots, x_{2n-1}}$$

Now since $R_{x_1, \dots, \hat{x}_i, \dots, x_{2n-1}} \in \text{Mult } J$ and $\text{Mult } J$ is generated by R_a where a is a word of length 1 or 2. We have

$$R_{x_1, \dots, \hat{x}_i, \dots, x_{2n-1}} = \sum_i \alpha_i R_{x_{i_1}}, \dots, R_{x_{i_{k_i}}}$$

And $k_i \geq \lceil \frac{2n-2}{2} \rceil = n - 1$. If there exist $x_i R_{x_{i_1}}, \dots, R_{x_{i_{k_i}}}$ where $k_i = n - 1$, then it is a word of the form:

$$x_i(a_1 b_1)(a_2 b_2) \dots (a_{n-1} b_{n-1})$$

Write it into operator on a_1 , then we get

$$x_i(a_1 b_1)(a_2 b_2) \dots (a_{n-1} b_{n-1}) = a_1 R_1 \dots R_n = 0$$

Therefore, if $(\text{Mult}(J))^n = 0$ then $J^{2n-1} = 0$. □

Exercise 1.7. Show that Peirce decomposition components satisfy the following relations:

$$J_{ii}^2 \subseteq J_{ii}, \quad J_{ij}^2 \subseteq J_{ii} + J_{jj}, \quad J_{ij} J_{jk} \subseteq J_{ik} \quad (i, j, k \text{ distinct}).$$

Proof. (1). Let $i \neq j$ and put $e = e_i + e_j$, $B = JU_e$. Then $B = J_{ii} + J_{ij} + J_{jj}$ is the Peirce decomposition of B relative to e_i, e_j . We have $J_{ii} = B_1(e_i)$, $J_{jj} = B_0(e_i) = J_{jj} = B_1(e_j)$, $J_{ij} = B_{1/2}(e_i) = J_{ji} = B_{1/2}(e_j)$. Hence $J_{ii}^2 = B_1^2(e_i) \subseteq B_1(e_i) = J_{ii}$ and $J_{ij}^2 = B_{1/2}^2(e_i) \subseteq B_1(e_i) + B_0(e_i) = J_{ii} + J_{jj}$.

(2). let i, j, k be distinct and put $f = e_i + e_j + e_k$, $C = JU_f$. Then $C = J_{ii} + J_{jj} + J_{kk} + J_{ij} + J_{ik} + J_{jk}$ is the Peirce decomposition of C relative to e_i, e_j, e_k ($J_{pq} = C_{pq}$). Also let $e = e_i + e_j$. Then e is an idempotent contained in C and $C_1(e) = J_{ii} + J_{ij} + J_{jj}$, $C_0(e) = J_{kk}$ and $C_{1/2}(e) = J_{ik} + J_{jk}$. Since $C_{1/2}(e)C_1(e) \subseteq C_{1/2}(e)$ we have $J_{ij}J_{jk} \subseteq J_{ik} + J_{jk}$ and since $J_{ij} = J_{ji}$, $J_{jk} = J_{kj}$ we have also $J_{ij}J_{jk} = J_{kj}J_{ji} \subseteq J_{ki} + J_{ji}$. Hence $J_{ij}J_{jk} \subseteq (J_{ik} + J_{jk}) \cap (J_{ki} + J_{ji}) = J_{ik}$. □

Exercise 1.8. Show that for any finite-dimensional Jordan algebra J , its nil-radical is hereditary, namely for any ideal I of J we have $\text{Nil } I = I \cap \text{Nil } J$.

Proof. ($\supseteq$) We have $I \cap \text{Nil } J \subseteq \text{Nil } I$ since $I \cap \text{Nil } J$ is a nil ideal of I .

($\subseteq$) For the other inclusion, consider $\bar{J} = J / \text{Nil } J$. Then we have $\text{Nil } \bar{J} = 0$. Let $\bar{I} = (I + \text{Nil } J) / \text{Nil } J$, which is an ideal of $\bar{J}$. By the previous corollary, since $\text{Nil } \bar{J} = 0$, we must have $\text{Nil } \bar{I} = 0$.

The canonical projection of $\text{Nil } I$ into $\bar{J}$ is $(\text{Nil } I + \text{Nil } J) / \text{Nil } J$. This is a nil ideal of $\bar{I}$. Since $\text{Nil } \bar{I} = 0$, we must have

$$(\text{Nil } I + \text{Nil } J) / \text{Nil } J = \{0\}$$

This implies $\text{Nil } I \subseteq \text{Nil } J$. Since by definition $\text{Nil } I \subseteq I$, we can conclude that $\text{Nil } I \subseteq I \cap \text{Nil } J$. □

Exercise 1.9. Prove that for the octonion algebra $\mathbb{O}$ (over real numbers), $\mathbb{O}^+$ is isomorphic to a Jordan algebra of a symmetric nondegenerate bilinear form.

Proof. □